Mars Rover Swarms

How Swarm Intelligence is Solving NASA's Biggest Challenges

AI IA Activity 1

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The Problem: A Lone Genius



NASA's traditional rovers, like *Perseverance*, are technological marvels. But they are also:

- **High-Cost:** Over $2.7 billion per mission.
- **High-Risk:** A single failure can end the mission.
- **High-Delay:** 22-minute lag makes real-time control impossible.
- **Limited:** Can't explore high-risk areas like caves or cliffs.

The Solution: Swarm Intelligence

Robust

Instead of one 'genius' robot, use 100 'good' robots. If 5 rovers fail, 95 keep working. No single point of failure.

Scalable

Can map vast areas, cover more ground, and perform multiple science tasks at the same time.

Flexible

Can explore high-risk zones like steep cliffs or lava tubes that a primary rover can't risk.

Case Study: The Research Paper

NASA's "Starling" Mission

To test this "group brain," NASA launched the Starling Mission in 2023.

It's a swarm of four CubeSats in Earth orbit, acting as a live test-bed for autonomous swarm software. This is the real-world research proving the concept.



How It Works: The "DSA" Brain

The key finding is the success of the **Distributed Spacecraft Autonomy (DSA)** software. This is the 'group mind' that lets the swarm operate without human control.

- **Autonomous Networking:** Satellites create their own communication network, deciding how to talk to each other.
- **Group Decision-Making:** The swarm can collectively analyze a situation and decide to perform a new task.
- **"Self-Healing" Network:** If one satellite's link is lost, the swarm automatically re-routes data around it.
- **Self-Management:** The DSA software manages its own power, memory, and data resources.

Key Findings & Results

100% Success

The DSA software achieved 100% success in autonomously managing the swarm's tasks, proving the core "group brain" concept is viable.

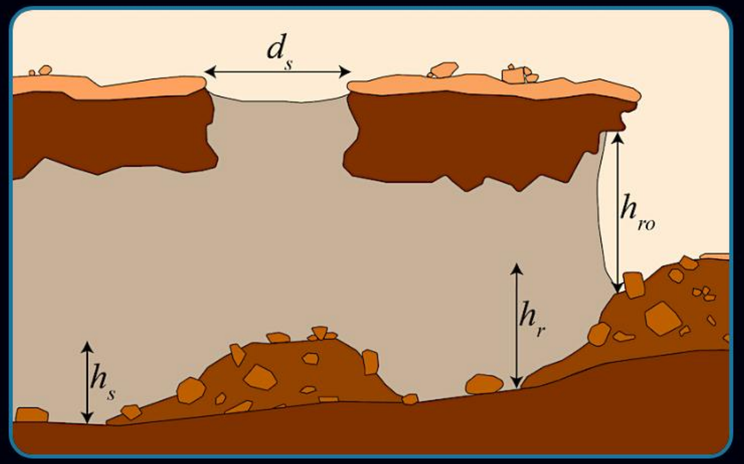
"Self-Healing"

The swarm demonstrated true "self-healing" by automatically re-routing data when connections were lost, proving its high resilience and robustness.

The Future: From Orbit to Mars



Aerial Swarms ("Marsbees")



Exploring Caves (Lava Tubes)



Rapid Area Mapping

References

National Aeronautics and Space Administration (NASA). (2025, February 4). *NASA Demonstrates Software 'Brains' Shared Across Satellite Swarms*. Retrieved from https://www.nasa.gov/centers-and-facilities/ames/nasa-demonstrates-software-brains-shared-across-satellite-swarms/

Thank You

Image Sources



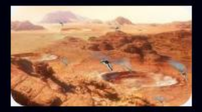
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